AGRIVIVE / BUND-ROOT

MADHYA PRADESH / CONTOUR-BUND INSPIRED / INFILTRATION CARTRIDGE

SERIES 13 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



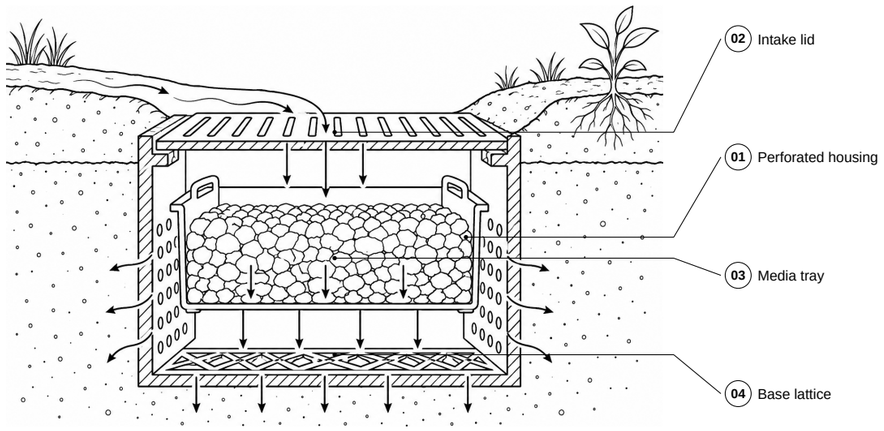
STATE CONTEXT / INDIGENOUS INSPIRATION

Contour bunding is studied for runoff and soil-moisture management in Madhya Pradesh. BUND-ROOT proposes a serviceable gravel cartridge in a contour strip; infiltration depends on the actual soil and rainfall.

Practice source: ICAR-IISWC: contour bunds in Datia, Madhya Pradesh | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 150 L x 80 W x 100 H

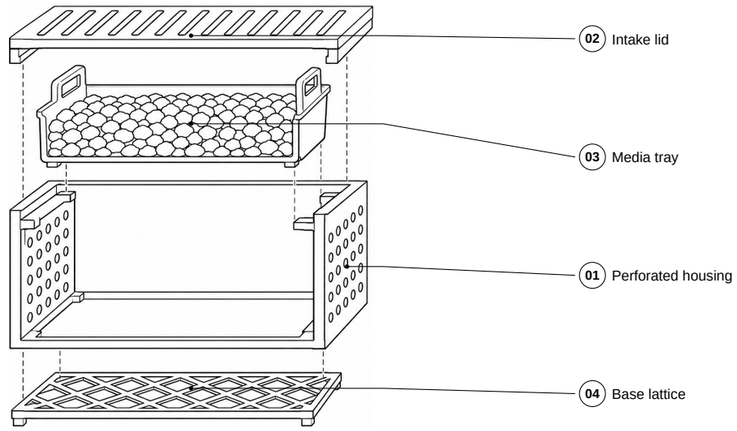


DESIGN / OPERATION NOTES

Seat housing beneath a shallow contour runoff path.
Fit the media tray with clean gravel and close the intake lid.
Water can pass through media and open side/base features into surrounding soil.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
150 L x 80 W x 100 H



DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

External L x W x H
150 x 80 x 100

Media compartment
120 x 60 x 60

Lid / slot detail
TBD

02 GEOMETRIC CHECK

GROSS MEDIA COMPARTMENT

432.0 mL

120 x 60 x 60 / 1000
Media void ratio determines actual water volume

03 PARTS / QTY

01 Perforated housing	**1**
02 Intake lid	**1**
03 Media tray	**1**
04 Base lattice	**1**

Tray/housing/lid/base: candidate polymer. Gravel charge: grading and void ratio TBD.

04 INTERFACES / CHECKS

Check intake clogging and soil migration.

Measure gravel void ratio and actual infiltration.

Check tray removal and cleaning clearance.

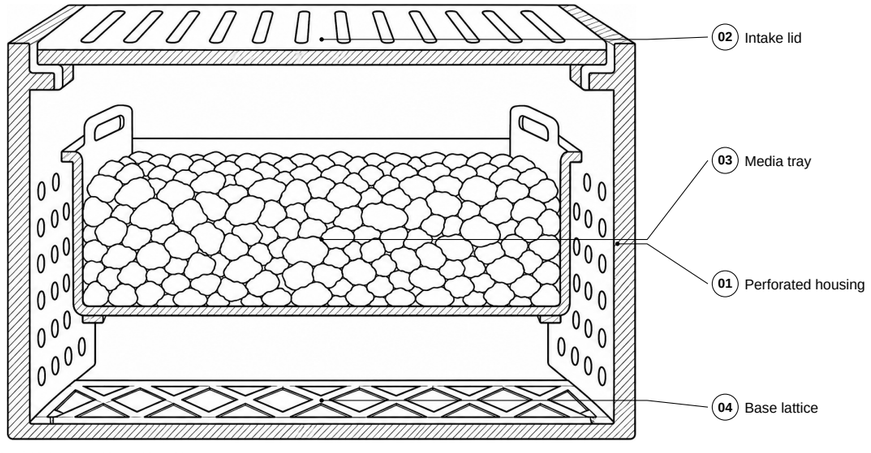
05 DATUMS / DRAWING BASIS

A: intake lid plane. B: base lattice. Keep side perforations open to soil.

PRELIMINARY / VALIDATION PENDING

04 INTERNAL SECTION

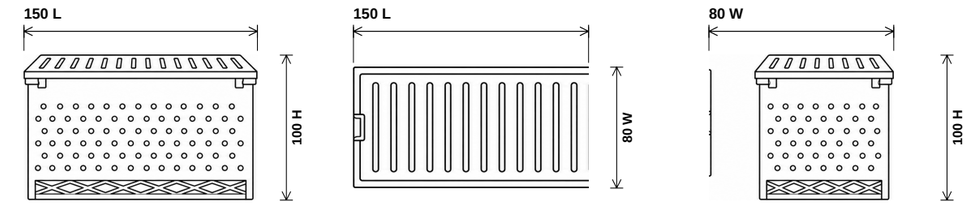
SECTION BASIS / mm
120 L x 60 W x 60 H media compartment



DESIGN / OPERATION NOTES

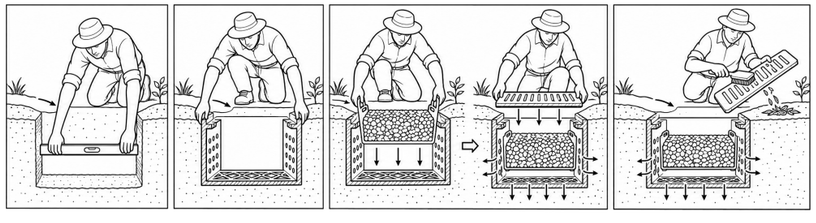
Fit the media tray with clean gravel and close the intake lid.
Water can pass through media and open side/base features into surrounding soil.

03 ORTHOGRAPHIC STUDIES / mm



External 150 L x 80 W x 100 H. Media envelope 120 L x 60 W x 60 H; not usable water storage.

05 INSTALLATION / OPERATING SEQUENCE



1 PREPARE STRIP **2 SEAT HOUSING** **3 FIT MEDIA / LID** **4 CLEAN INTAKE**

Prototype fit, access and operation require trials.

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-13 / P01	BUND-ROOT	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	13 / 28